

Decision Tree Classification: Example Problem

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Entropy and Information Gain

$$\text{Entropy}(p) = - \sum_i p_i \log_2 p_i, \quad \text{Gain}(S, F) = \text{Entropy}(S) - \sum_{f \in \text{values}(F)} \frac{|S_f|}{|S|} \text{Entropy}(S_f).$$

As an example, suppose that we have data (with outcomes) $S = \{s_1 = \text{true}, s_2 = \text{false}, s_3 = \text{false}, s_4 = \text{false}\}$ and one feature F that can have values $\{f_1, f_2, f_3\}$. In the example, the feature value for s_1 could be f_2 , for s_2 it could be f_2 , for s_3 , f_3 and for s_4 , f_1 then we can calculate the entropy of S as (where $\oplus$ means true, of which we have one example, and $\ominus$ means false, of which we have three examples):

$$\begin{aligned} \text{Entropy}(S) &= -p_{\oplus} \log_2 p_{\oplus} - p_{\ominus} \log_2 p_{\ominus} \\ &= -\frac{1}{4} \log_2 \frac{1}{4} - \frac{3}{4} \log_2 \frac{3}{4} \\ &= 0.5 + 0.311 = 0.811. \end{aligned} \tag{12.3}$$

Entropy and Information Gain

$$\frac{|S_{f_1}|}{|S|} \text{Entropy}(S_{f_1}) = \frac{1}{4} \times \left(-\frac{0}{1} \log_2 \frac{0}{1} - \frac{1}{1} \log_2 \frac{1}{1} \right)$$

$$= 0$$

$$\frac{|S_{f_2}|}{|S|} \text{Entropy}(S_{f_2}) = \frac{2}{4} \times \left(-\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} \right)$$

$$= \frac{1}{2}$$

$$\frac{|S_{f_3}|}{|S|} \text{Entropy}(S_{f_3}) = \frac{1}{4} \times \left(-\frac{0}{1} \log_2 \frac{0}{1} - \frac{1}{1} \log_2 \frac{1}{1} \right)$$

$$= 0$$

The information gain from adding this feature is the entropy of S minus the sum of the three values above:

$$\text{Gain}(S, F) = 0.811 - (0 + 0.5 + 0) = 0.311. \quad (12.7)$$

ID3 Algorithm

The ID3 Algorithm

- If all examples have the same label:
 - return a leaf with that label
 - Else if there are no features left to test:
 - return a leaf with the most common label
 - Else:
 - choose the feature $\hat{F}$ that maximises the information gain of S to be the next node using Equation (12.2)
 - add a branch from the node for each possible value f in $\hat{F}$
 - for each branch:
 - * calculate S_f by removing $\hat{F}$ from the set of features
 - * recursively call the algorithm with S_f , to compute the gain relative to the current set of examples
-

The computational cost of constructing binary trees is well known for the general case, being $O(N \log N)$ for construction and $O(\log N)$ for returning a particular leaf, where N is the number of nodes

Decision Tree Algorithm

Decision tree construction

Algorithm: DecisionTree (D, F)

input : Data set D , features F

output: Feature tree T with labelled leaves

```
1 if Homogeneous( $D$ ) then return Label( $D$ )
2  $f \leftarrow$  BestFeature ( $D, F$ )
3 Partition  $D$  into  $\{D_1 \dots D_L\}$  based on the literals  $f = v$ 
4 foreach  $D_i$  do
5     if  $D_i \neq \emptyset$  then  $T_i \leftarrow$  DecisionTree ( $D_i, F$ )
6     else  $T_i$  is a leaf labelled with Label ( $D$ )
7 end
8 return a tree whose root is labelled with  $f$  and whose
   children are  $T_i$ 
```

Decision Tree Algorithm

Best feature for partitioning the data set

Algorithm: BestFeature (D, F)

input : Data set D , features F

output: Best feature f to partition D

```
1 Impuritymin  $\leftarrow$  1
2 foreach  $f \in F$  do
3    $\{D_1 \dots D_L\} \leftarrow$  Partition  $D$  based on  $f$  values.
4   if Impurity ( $\{D_1 \dots D_L\}$ )  $<$  Impuritymin then
5     Impuritymin  $\leftarrow$  Impurity  $\{D_1 \dots D_L\}$ 
6      $f_{best} \leftarrow f$ 
7   end
8 end
9 return  $f_{best}$ 
```

Decision Tree – Classification Example

Deadline?	Is there a party?	Lazy?	Activity
Urgent	Yes	Yes	Party
Urgent	No	Yes	Study
Near	Yes	Yes	Party
None	Yes	No	Party
None	No	Yes	Pub
None	Yes	No	Party
Near	No	No	Study
Near	No	Yes	TV
Near	Yes	Yes	Party
Urgent	No	No	Study

$$\begin{aligned}\text{Entropy}(S) &= -p_{\text{party}} \log_2 p_{\text{party}} - p_{\text{study}} \log_2 p_{\text{study}} \\ &\quad - p_{\text{pub}} \log_2 p_{\text{pub}} - p_{\text{TV}} \log_2 p_{\text{TV}} \\ &= -\frac{5}{10} \log_2 \frac{5}{10} - \frac{3}{10} \log_2 \frac{3}{10} - \frac{1}{10} \log_2 \frac{1}{10} - \frac{1}{10} \log_2 \frac{1}{10} \\ &= 0.5 + 0.5211 + 0.3322 + 0.3322 = 1.6855\end{aligned}$$

Decision Tree – Classification Example

$$\begin{aligned}
 \text{Gain}(S, \text{Deadline}) &= 1.6855 - \frac{|S_{\text{urgent}}|}{10} \text{Entropy}(S_{\text{urgent}}) \\
 &\quad - \frac{|S_{\text{near}}|}{10} \text{Entropy}(S_{\text{near}}) - \frac{|S_{\text{none}}|}{10} \text{Entropy}(S_{\text{none}}) \\
 &= 1.6855 - \frac{3}{10} \left(-\frac{2}{3} \log_2 \frac{2}{3} - \frac{1}{3} \log_2 \frac{1}{3} \right) \\
 &\quad - \frac{4}{10} \left(-\frac{2}{4} \log_2 \frac{2}{4} - \frac{1}{4} \log_2 \frac{1}{4} - \frac{1}{4} \log_2 \frac{1}{4} \right) \\
 &\quad - \frac{3}{10} \left(-\frac{1}{3} \log_2 \frac{1}{3} - \frac{2}{3} \log_2 \frac{2}{3} \right) \\
 &= 1.6855 - 0.2755 - 0.6 - 0.2755 \\
 &= 0.5345
 \end{aligned}$$

Deadline?	Is there a party?	Lazy?	Activity
Urgent	Yes	Yes	Party
Urgent	No	Yes	Study
Near	Yes	Yes	Party
None	Yes	No	Party
None	No	Yes	Pub
None	Yes	No	Party
Near	No	No	Study
Near	No	Yes	TV
Near	Yes	Yes	Party
Urgent	No	No	Study

$$\begin{aligned}
 \text{Gain}(S, \text{Party}) &= 1.6855 - \frac{5}{10} \left(-\frac{5}{5} \log_2 \frac{5}{5} \right) \\
 &\quad - \frac{5}{10} \left(-\frac{3}{5} \log_2 \frac{3}{5} - \frac{1}{5} \log_2 \frac{1}{5} - \frac{1}{5} \log_2 \frac{1}{5} \right) \\
 &= 1.6855 - 0 - 0.6855 \\
 &= 1.0
 \end{aligned}$$

$$\begin{aligned}
 \text{Entropy}(S) &= -p_{\text{party}} \log_2 p_{\text{party}} - p_{\text{study}} \log_2 p_{\text{study}} \\
 &\quad - p_{\text{pub}} \log_2 p_{\text{pub}} - p_{\text{TV}} \log_2 p_{\text{TV}} \\
 &= -\frac{5}{10} \log_2 \frac{5}{10} - \frac{3}{10} \log_2 \frac{3}{10} - \frac{1}{10} \log_2 \frac{1}{10} - \frac{1}{10} \log_2 \frac{1}{10} \\
 &= 0.5 + 0.5211 + 0.3322 + 0.3322 = 1.6855
 \end{aligned}$$

$$\begin{aligned}
 \text{Gain}(S, \text{Lazy}) &= 1.6855 - \frac{6}{10} \left(-\frac{3}{6} \log_2 \frac{3}{6} - \frac{1}{6} \log_2 \frac{1}{6} - \frac{1}{6} \log_2 \frac{1}{6} - \frac{1}{6} \log_2 \frac{1}{6} \right) \\
 &\quad - \frac{4}{10} \left(-\frac{2}{4} \log_2 \frac{2}{4} - \frac{2}{4} \log_2 \frac{2}{4} \right) \\
 &= 1.6855 - 1.0755 - 0.4 \\
 &= 0.21
 \end{aligned}$$

Decision Tree – Classification Example

Deadline?	Is there a party?	Lazy?	Activity
Urgent	Yes	Yes	Party
Urgent	No	Yes	Study
Near	Yes	Yes	Party
None	Yes	No	Party
None	No	Yes	Pub
None	Yes	No	Party
Near	No	No	Study
Near	No	Yes	TV
Near	Yes	Yes	Party
Urgent	No	No	Study

Deadline?	Is there a party?	Lazy?	Activity
Urgent	No	Yes	Study
None	No	Yes	Pub
Near	No	No	Study
Near	No	Yes	TV
Urgent	No	Yes	Study

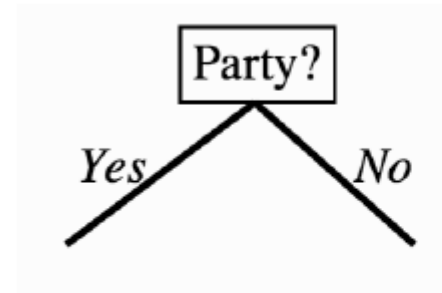


FIGURE 12.6 The decision tree after one step of the algorithm.

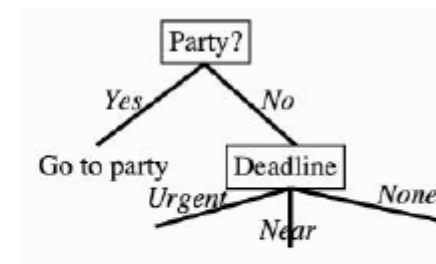
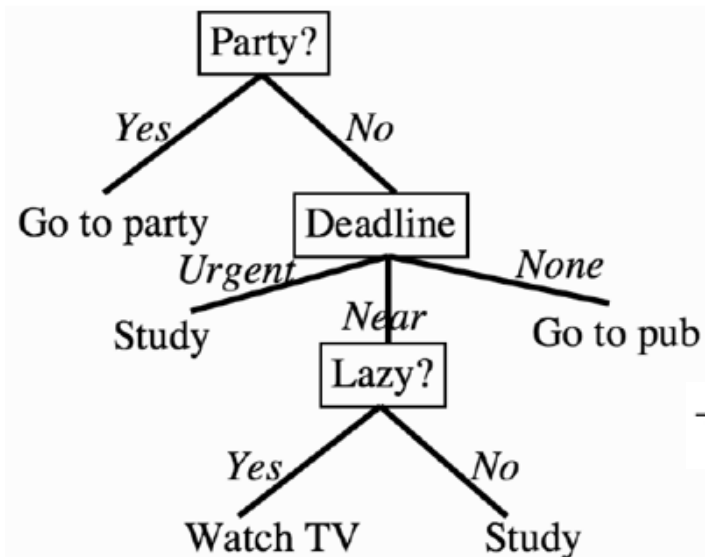


FIGURE 12.7 The tree after another step.

Decision Tree – Classification Example

Deadline?	Is there a party?	Lazy?	Activity
Urgent	Yes	Yes	Party
Urgent	No	Yes	Study
Near	Yes	Yes	Party
None	Yes	No	Party
None	No	Yes	Pub
None	Yes	No	Party
Near	No	No	Study
Near	No	Yes	TV
Near	Yes	Yes	Party
Urgent	No	No	Study

Deadline?	Is there a party?	Lazy?	Activity
Urgent	No	Yes	Study
None	No	Yes	Pub
Near	No	No	Study
Near	No	Yes	TV
Urgent	No	Yes	Study

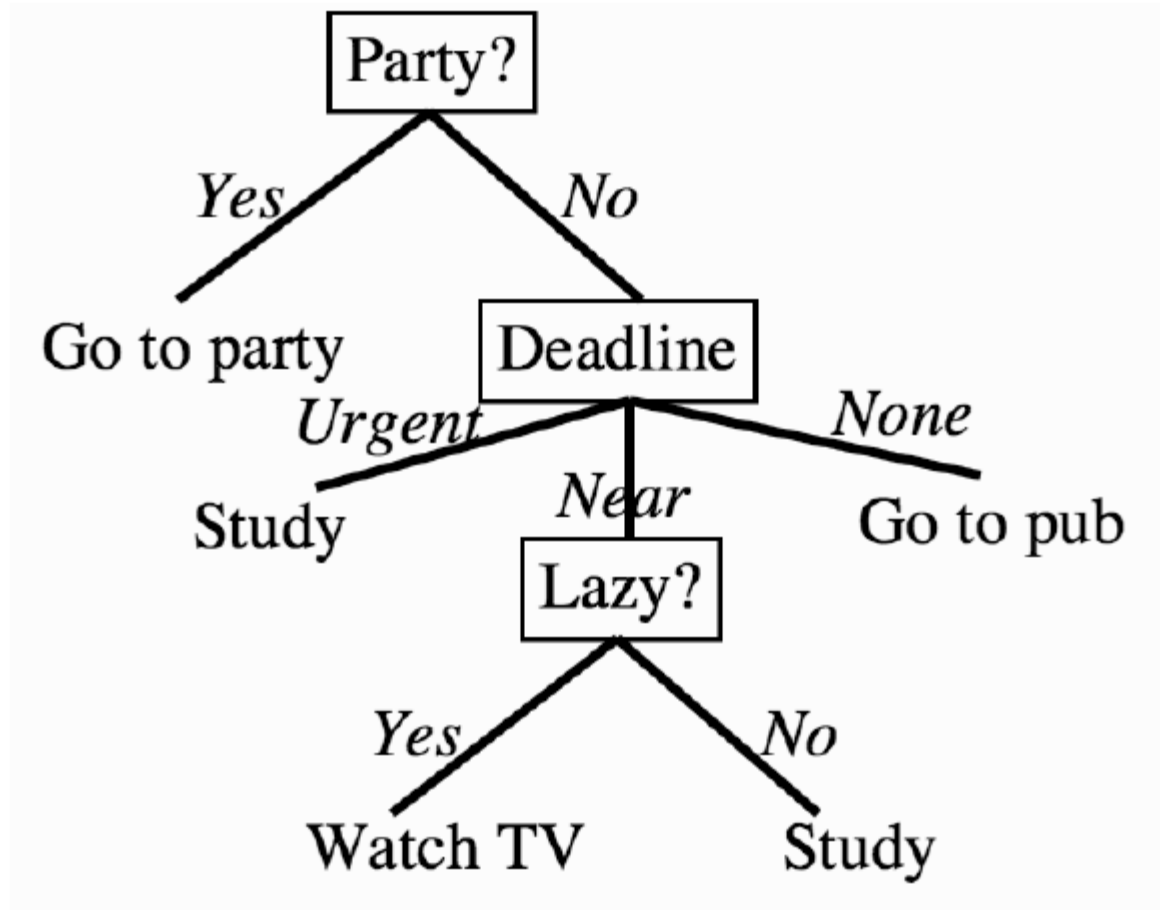


Deadline?	Is there a party?	Lazy?	Activity
Near	No	No	Study
Near	No	Yes	TV

$$\begin{aligned}
 \text{Gain}(S, \text{Deadline}) &= 1.371 - \frac{2}{5} \left(-\frac{2}{2} \log_2 \frac{2}{2} \right) \\
 &\quad - \frac{2}{5} \left(-\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} \right) - \frac{1}{5} \left(-\frac{1}{1} \log_2 \frac{1}{1} \right) \\
 &= 1.371 - 0 - 0.4 - 0 \\
 &= 0.971
 \end{aligned}$$

$$\begin{aligned}
 \text{Gain}(S, \text{Lazy}) &= 1.371 - \frac{4}{5} \left(-\frac{2}{4} \log_2 \frac{2}{4} - \frac{1}{4} \log_2 \frac{1}{4} - \frac{1}{4} \log_2 \frac{1}{4} \right) \\
 &\quad - \frac{1}{5} \left(-\frac{1}{1} \log_2 \frac{1}{1} \right) \\
 &= 1.371 - 1.2 - 0 \\
 &= 0.1710
 \end{aligned}$$

A Simple Decision Tree



- . A simple decision tree to decide how you will spend the evening.

Workout Problem

Problem 12.3 Turn this politically incorrect data from Quinlan into a decision tree to classify which attributes make a person attractive, and then extract the rules.

Height	Hair	Eyes	Attractive?
Small	Blonde	Brown	No
Tall	Dark	Brown	No
Tall	Blonde	Blue	Yes
Tall	Dark	Blue	No
Small	Dark	Blue	No
Tall	Red	Blue	Yes
Tall	Blonde	Brown	No
Small	Blonde	Blue	Yes

Reference

- Stephen Marsland, “Machine Learning – An Algorithmic Perspective”, 2nd Edition, Chapman and Hall/CRC Machine Learning and Pattern Recognition Series, 2015